

A QUANTITATIVE APPROACH TO THE MARSUPIAL BRAIN IN AN ECO-ETHOLOGICAL PERSPECTIVE¹

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ABSTRACT — Using a volumetric method and the classical allometry equation, the author investigates the possibility of a relationship between the size of total brain or brain-components and the ecology or ethology of 29 marsupials species from South-America and Australia. He shows that there is a relation between brain and neocortex volumes, and the taxonomic groups characterized by definite life-habits. New light is shed upon the evolution of marsupials showing that, while the living didelphids most probably are the closest to the ancestral type, some other groups such as the dasyurids and peramelids include even more primitive forms. This points to a very early separation of American and Australian marsupials and to a long independent evolution, perhaps dating back to immediately pre-marsupial times.

This paper is an approach to a study of general brain organization in marsupials in relation to their life-habits. It is based on the same principles as previous reports on bats by Pirlot and other researchers. Its scope is to investigate the possibility of some relationship between quantitative brain-organization and life-habits in marsupials, as broadly defined by both diet and locomotion. Thus, this work falls into the field of functional morphology. The discussion will contain an examination of the current views on the evolution of the sub-class; however, one should not anticipate too much in that respect.

The sub-class Marsupialia is especially interesting from the viewpoint just stated above. Firstly, it is a

relatively low-ranking group, along with the Monotremata, that is to say relatively close to the mammalian ancestor or ancestors. It thus represents an important stepping-stone in the evolution from reptiles to mammals. Secondly, it shares with the order Chiroptera one attractive peculiarity. While showing an indisputable unity (or homogeneity) in its main characteristic i.e. its reproduction pattern, it also offers a very broad spectrum of morphological and functional (behavioural) features i.e. a variety (or heterogeneity) of diets and locomotions. The diets are either insectivorous or carnivorous or both combined, or vegetarian (browsing or grazing type), sometimes frugivorous. Locomotion includes walking, tree climbing and jumping, gliding, digging, hopping and swimming.

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TABLE I

List of material

	Species	Collection	Sex	Use	Body-weight gr	Brain-weight mgr
Polyprotodonta						
Didelphidae						
(1)	<i>Didelphis m. marsupialis</i>	P313	F	S	1205	6885
(2)	<i>Didelphis m. aurita</i>	P321	F	S	1150	6033
(3)	<i>Didelphis azarae</i>	J65069	M	W	1229	5100
(4)	<i>Didelphis virginiana</i>	J66073	F	W	1474	6420
(5)	idem	J66075	M	S	2494	9300
(6)	<i>Philander opossum</i>	J66076	F	S	594	4330
(7)	idem	J66077	M	W	558	4150
(8)	<i>Metachirops (?) sp.</i>	P326	F	S	375	3305
(9)	<i>Marmosa (Marmosa) murina</i>	J65067	M	S	22	770
(10)	<i>Marmosa (m.) domina</i>	P327	M	S	100	1403
(11)	<i>Marmosa (Thylamys) sp.</i>	J70092	?	S	170	(2200)
(12)	<i>Caluromys lanatus</i>	J66074	M	W	378	4650
(13)	idem	J66071	M	W	501	4900
(14)	idem	J66070	M	S	418	4540
Dasyuridae						
(15)	<i>Antechinus flavipes</i> (x)	J65047	F	S	32	760
(16)	idem (x)	J65048	F	W	36	770
(17)	<i>Sminthopsis murina</i>	J64017	M	S	10	440
(18)	<i>Dasyurus quoll</i>	J65060	F	S	865	7260
(19)	idem	J65061	M	W	1162	6620
(20)	idem	J65062	F	W	709	6900
(21)	<i>Sarcophilus harrisii</i>	J65063	F	W	3742	14500
(22)	idem	J65065	F	S	5060	16260
Peramelidae						
(23)	<i>Isoodon obesulus</i>	J64031	F	S	1120	5260
(24)	idem	J65032	M	W	1372	5310
(25)	idem	J65066	F	W	1570	5150
(26)	<i>Perameles nasuta</i>	J65043	M	S	1050	5880
(27)	idem	J65049	F	W	1020	5700
Paucituberculata						
Caenolestidae						
(28)	<i>Caenolestes obscurus</i>	J69087	M	S	26.5	979
(29)	<i>Lestoros inca</i>	J70094	?	S	25.8	790
Diprotodonta						
Petauridae						
(30)	<i>Petaurus breviceps</i>	J64020	M	S	171	2480
(31)	idem	J65053	F	W	110	2390
(32)	<i>Petaurus norfolcensis</i>	J64028	F	W	250	3230
(33)	<i>Schoinobates volans</i>	J64037	M	S	1305	4910
(34)	<i>Pseudocheirus peregrinus</i>	J64019	M	W	658	5520
(35)	idem	J64026	F	W	476	4720
(36)	idem	J64038	F	S	728	5570

TABLE I (suite)

List of material

Species	Collection	Sex	Use	Body-weight gr	Brain-weight mgr
<i>Phalangeridae</i>					
(37) <i>Trichosurus vulpecula</i>	J64029	F	W	1040	12120
(38) idem	J64040	F	S	2150	12970
<i>Vombatidae</i>					
(39) <i>Vombatus mitchelli</i> (ursinus?)	J64011	M	S	15876	55400
<i>Macropodidae</i>					
(40) <i>Potorous tridactylus</i>	J65055	M	S	1304	12340
(41) idem	J65056	M	W	1358	12130
(42) <i>Setonix brachyurus</i>	J64035	M	W	2160	17520
(43) idem	J64036	F	W	2220	12780
(44) idem	J67079	M	S	3170	14210
(45) idem	J67080	F	W	2430	14100
(46) <i>Thylogale billardieri</i>	J65057	M	S	8845	25720
(47) <i>Macropus eugenii</i>	J64033	F	S	3580	12560
(48) idem	J64034	M	W	3640	16150
(49) <i>Macropus rufogriseus</i>	J65064	M	S	12786	36560
(50) <i>Macropus giganteus</i>	P309	F	S	21000	53850

S = sectioned and used for component analysis as well as for encephalization estimate.
W = weighed and used for encephalization estimate only. The *Antechinus* are marked with (x): the collector noted that he cannot say whether they were adult or young. Specimen (8), *Metachirops* raises a problem that I cannot discuss here. That generic name is now made synonymous with *Philander*. My specimen was fairly different from the usual *Philander* type, and more like Cabrera and Yezpez' *Metachirops* and a Brazilian zoologist told me he considered it *Metachirops*. My specimen was rather light but an adult female with 4 healthy young in the pouch.

All the arguments formerly presented to stress the significance of the brain of bats for the neurobiologist oriented towards a study of macro-evolution apply to the brain of marsupials. Further, the information I have to-day on the behaviour of marsupials in their various environments is about the same as the one I had when I first became interested in the possible relationships between the quantitative features of the brain of bats and their ecology and ethology (Pirlot, 1969, 1970). That introductory work led to more papers along the same lines in collaboration with other researchers (Pirlot and Stephan, 1970 Stephan and

Pirlot, 1970; Stephan, Pirlot and Schneider, 1974; Pirlot and Pottier, 1977) using a comparative method devised by Stephan (1967) and applied by him and his co-authors to the investigation of the brains of Insectivores and Primates.

This article was inspired by a verbal suggestion by Prof. John Nelson, Monash University, Clayton, Va, Australia, that the concepts and methods that had been used in our work on bats could also be applied to marsupials. A first approach to the encephalization in marsupials, based on published data but not using those methods, can be found in Moeller (1973).

MATERIALS AND METHOD

A complete list of the individual animals studied is given in Table I. Two very unequal collections were involved. One is designated by the initial P followed by a registration number. This designates «Pirlot's collection» at the Université de Montréal, Montréal, Canada. It contains five specimens only, one of which (P 321) was supplied by Prof. E. Oswaldo-Cruz and Prof. C. Rocha-Miranda, Universidade Federal do Rio de Janeiro, Brazil, and one (P 309) by Dr. J. Nelson. The other (45) animals belong to the collection of Prof. J. I. Johnson, Michigan State University, East Lansing, USA.

The brains were fixed in formalin and embedded in paraffin or celloidin. They were sliced at 10 to 40 μ m and stained for Nissl substance. The body-weights and brain-weights were supplied by the collectors. Brain density (needed to transform weights into volumes) was assumed to be equal, on the average, to that of placental brains: 1.036.

In all, three living orders contribute to the study material. The first one, the Polyprotodonta, is represented by three families; the second one, the Paucituberculata, includes one family; the third one, the Diprotodonta, is represented by four families. The total includes 21 genera and 29 species. The diets are predominantly either insectivorous or carnivorous (or both combined) or vegetarian (browsing or grazing type), sometimes frugivorous. Locomotion includes walking, tree climbing and jumping, gliding, digging, hopping; the significance of swimming is unknown to me.

The method is quantitative and comparative. It has been described, perhaps too briefly and not always without some conceptual confusion, in various papers. It involves three steps. First, simple size ratios BrW/BoW (BrW = brain-weight; BoW = body-weight) are used as an indication of the morpho-functional relationship between the central nervous system which has the controlling power, and the body which is being controlled. Secondly, the allometric regression of total brain-weight on body-weight is investigated. This relationship is expressed by the classical equation $y = bx^k$ where y = brain-weight, x = body-weight, b = a constant and k = the growth-rate. It is the basis for the concept of encephalization. Finally, the allometric regressions of brain components (by volumes) on body-weights are considered. The same type of equation is used, y here desi-

gnating the volume of the brain component. This relationship allows for further insight into the functional organization of the brain by drawing attention to the high or low development of specific brain parts that are related either to general abilities (e.g. the neocortex for association or versatility) or to special skills (e.g. the bulbus olfactorius for smell).

One particular concern in this work is to throw some light on the evolutionary, or phylogenetic, picture of the marsupials as seen from the viewpoint of quantitative brain development. The best method available at present certainly is Stephan's (1967) progression index. This involves allometrical comparisons with a reference group taken as a standard (basal) type, and chosen as the closest possible approximation to the hypothetical extinct ancestor(s) of the living animals under study. The principle of this approach has been re-explained several times (e.g. Pirlot and Stephan, 1970; Stephan and Pirlot, 1970), and it is simple: the progression index indicates how many times the brain (or brain component) of a living form is larger or smaller than the brain (or brain component) of the basal reference type of equal body-weight. Multiplied by 100, those ratios make ascending or descending scales in which every figure above 100 indicates progression and every figure below 100 means regression with respect to the basal level of 100.

For reasons mainly related to paleontological data, the basal group chosen here is composed of a set of eight species or subspecies of Didelphinae. The rationale for that choice is presented in Appendix A. The Didelphinae in my sample vary in absolute size from 22 to 2494 gr. i.e. over a hundredfold range (specimens # 9 and 5 in Table I). That basal group yielded regression equations for total brain and for each brain component, in the form $Y = B + kX$ where $Y = \log y$, $B = \log b$ and $X = \log x$ of the equation given above. Since constants B and k were found by the least square method for the Didelphinae, it is easy to take as X the log of the body-weight of any other marsupial to obtain Y , the log of the brain-weight (or brain component), hence the brain-weight or the brain-component volume of a basal didelphine theoretically having that body-weight. Then, the ratio of the actual brain-weight or brain-component volume to that hypothetical basal weight or volume is by definition the progression index (P.I. = observed actual/expected basal).

It is easy to estimate the brain volume from the density factor applied to the brain weight and this leads to brain component volumes which are calculated through the tedious but straightforward technique of photographing microscopical sections, as explained, for example, in Stephan, Bauchot and Andy (1970). Here, the number of sections used per brain varied between 115 and 130. The old technique of clipping the neuronal fields from photographs then weighing them was preferred to the use of an electronic area analyzer because it allows for verification which is not possible with the currently available electronic instrument. Some problems arose in delimiting certain brain-components. Comments on this are given in Appendix B.

The following abbreviations for brain-components will be used throughout this paper. BO = bulbus olfactorius; N = neocortex; RH = rhinencephalon; PY = pyriform cortex; S = septum; Di = diencephalon; STR = striatum; Hi = hippocampus; M = mesencephalon; C = cerebellum; O = medulla oblongata; Sz = schizocortex.

RESULTS

The results are presented first for the basal Didelphinae, then for the other species.

BASAL DIDELPHINAE

1) *Simple BrW/BoW ratios.* As observed in other mammals, the brain-to-body ratio in this sample increases as absolute size decreases (from approximately 0.44% to 3.5%).

2) *Basal BrW/BoW regression.* After interpolating the BrW for specimen # 11 (*Marmosa Thyblamys*), one obtains the log-regression $Y = 2.1446 + 0.5315 X$, with $r = 0.9917$. This is the basal regression for encephalization. It may be of interest that the basal regression for placentals calculated by Stephan from basal Insectivores is $Y = 1.632 + 0.63 X$ (Pirlot and Stephan, 1970). The didelphine line thus has a somewhat lower slope.

3) *Simple component-to-brain ratios.* Nine brains from the Didelphidae were sectioned and analyzed by volume (Table II). The percentage of each component in total brain for the same group is given in Table III. *Caluromys* is listed separately. The pooled olfactory structures (BO + RH + PY) in *Caluromys* yield a clearly lower ratio than in the other didelphine species (16.44% in contrast to an average of 23.86%) whereas the ratio for the diencephalon is 12.29% in *Caluromys* compared with an average of 9.00% in the other species. These contrasting figures, very useful as a preliminary exploration, reinforce the validity of separating *Caluromys* from the other didelphids, as has been suggested by some taxonomists on other morphological grounds. (Reig, in Kirsch, 1977). More on this point is found in Appendix A.

4) *Basal regressions for brain components.* For the eight Didelphinae selected (excluding *Caluromys*) and sectioned, basal equations were calculated for thirteen components. Their constants are shown in Table IV.

NON-BASAL SPECIES

1) *Simple BrW/BoW ratios.* The ratios vary from 0.25% to 4.4%. An average was taken whenever a species was represented by more than one individual. While the general rule, the increase in relative brain-weight with decrease in body-weight, is clearly followed in that sample, there is a wide variability of ratios, a fact that researchers sometimes underestimated (see Appendix C).

2) *Progression of total brains.* The progression indices estimated by the method indicated above for total brains are given in Table V.

TABLE II

This table presents the actual volumes in cubic mm for 12 brain components and their sum making up total brain for each species. The last line is the sum of RH and PY which is equivalent to RH in my work on bats in which PY was not considered separately. The figures for the first 8 species (from left) are those used to estimate the basal (didelphine) regressions

Polyprotodonta															
Didelphidae									Peramelidae			Dasyuridae			
Didelphinae									Microbiotheriinae	Peramelinae		Phascogalinae		Dasyurinae	
Volum- es	Mar- mosa murina	Mar- mosa domina	Mar- mosa thyla- mys	Meta- chirops	Phi- lander	Didel- phis aurita	Didel- phis marsu- pialis	Didel- phis virgi- niana	Calu- romys	Pera- meles	Isoden	Ante- chinus	Smin- thopsis	Dasy- urus	Sarco- philus
BO	39.5	104.4	141.2	203.1	341.3	372.7	581.3	759.9	180.8	490.1	425.2	53.6	31.5	461.7	1136.0
N	176.5	290.0	448.8	759.7	904.0	1199.1	1446.2	1681.0	965.1	913.7	853.9	118.2	86.4	1773.6	3733.4
RH	68.6	146.0	225.8	385.2	418.9	593.5	805.6	1043.6	342.9	593.5	511.5	81.3	46.7	675.5	1605.5
PY	34.1	60.1	125.8	141.6	235.9	269.5	378.0	470.3	177.3	386.1	287.1	40.9	22.3	310.3	669.7
S	7.8	10.4	12.2	24.1	30.7	38.9	60.8	74.9	40.1	46.9	65.2	6.1	3.4	48.6	102.0
Di	61.3	122.4	167.6	265.9	341.1	573.0	638.5	809.9	527.5	471.2	427.3	63.4	38.7	625.9	1472.5
STR	24.6	33.3	52.9	72.7	109.4	122.7	202.3	213.4	110.4	163.4	149.8	23.8	12.9	197.2	493.4
Hi	63.2	108.9	200.2	257.7	373.9	480.3	536.4	620.5	360.3	579.7	578.2	57.7	39.2	453.0	823.3
M	71.1	124.2	206.5	197.9	295.5	428.9	379.0	561.3	350.1	320.1	321.6	74.8	36.6	447.5	923.8
C	98.3	166.2	273.7	412.4	563.4	916.9	852.6	1407.9	686.6	874.7	743.4	92.7	50.2	1100.2	2500.5
O	83.2	166.9	176.0	346.3	402.0	651.3	66.0	844.4	498.5	567.9	504.7	96.2	50.5	660.9	1519.6
Sz	4.2	8.4	34.6	22.8	33.9	111.3	38.3	40.1	54.5	77.5	79.4	5.9	3.5	69.1	172.0
	732.4	1341.2	2065.3	3089.4	4050.0	5758.1	6485.0	8527.2	4295.1	5484.8	4947.3	714.6	421.9	6823.5	15151.7
RH PY }	102.7	206.1	351.6	526.8	654.8	863.0	1183.6	1513.9	520.2	979.6	798.6	122.2	69.0	985.8	2275.2

Diprotodonta											
Volum- es	Phalangeridae				Macropodidae						Vombatidae
	Phalangerinae	Phascolarctinae	Trichosurinae	Potoroinae	Macropodinae						Vombatus
	Petaurus	Schoino- bates	Pseudo- cheirus	Tricho- surus	Potorus	Macro- pus Eugenii	Setonix	Thylo- gale	Macro- pus rufo- griseus	Macro- pus gigon- teus	
Bo	57.5	113.8	95.5	308.4	635.0	308.8	289.2	646.8	555.3	598.6	1378.0
N	735.2	1421.3	1639.4	4440.9	3875.5	4398.7	4600.2	9950.7	14284.9	23932.6	24131.0
RH	179.5	242.0	373.0	961.3	1090.1	746.6	1045.4	1672.7	1823.1	3538.7	3229.9
PY	87.8	76.4	108.3	262.7	452.6	362.4	341.1	446.0	360.7	683.4	1282.5
S	15.7	31.7	42.8	65.7	81.6	83.7	80.1	99.3	111.5	242.1	293.4
Di	316.5	517.9	574.1	1250.9	1124.4	1350.4	1506.5	2338.9	3188.9	5768.6	5236.0
STR	97.9	185.0	144.9	527.2	381.8	361.3	386.9	734.1	740.5	1477.6	1929.7
HI	157.4	254.1	460.4	671.1	1296.6	967.7	1123.4	2004.6	2261.6	2874.1	2887.9
M	110.7	319.5	339.4	509.3	484.0	533.2	792.5	980.7	1468.0	1523.0	1433.3
C	357.4	903.9	881.0	1844.4	1435.9	1632.5	2022.3	3577.6	6281.4	6883.9	7094.6
O	230.4	482.0	588.3	1181.3	784.0	824.0	1225.3	1830.8	3295.1	3508.2	3086.3
Sz	28.3	56.7	57.2	112.0	107.4	94.4	87.4	137.5	199.8	451.5	633.1
	2374.3	4604.3	5304.3	12135.2	11748.9	11663.7	13500.3	24419.7	34570.8	51482.3	52645.7
RH PY }	267.3	318.4	481.3	1224.0	1542.7	1109.0	1386.5	2118.7	2183.8	4222.1	4512.4

Paucituberculata		
Volumes	Caenolestidae	
	Lestoros	Caenolestes
Bo	77.4	83.2
N	132.3	184.2
RH	96.0	119.8
PY	55.0	72.0
S	6.4	6.6
D	56.9	74.3
ST	29.3	26.9
H	61.0	86.1
M	53.7	52.2
C	104.0	129.5
O	81.5	91.0
Sz	2.9	3.7
	756.4	929.5
RH } PY }	151.0	191.8

3) Progression of brain-components.

The progression indices are shown in Table VI. It will be noticed that the combination RH + PY has been treated as an entity with its distinct constant so that it is equivalent to Rh alone in other studies, especially those on bats in which the authors have applied a similar analysis. (Stephan and Pirlot, 1970; Pirlot and Pottier, 1977).

DISCUSSION

A. ENCEPHALIZATION

For the 21 marsupial species referred to the basal or standard 8 Didelphinae, the encephalization indices are comparable to those obtained for certain placentals of relatively low systematic ranks in relation to the basal Insectivores. They range from 78 to 232 for the 21 marsupials and from 70 to 275 among Microchiroptera (Pirlot and Pottier, 1977), reaching 382 in Megachiroptera (Stephan, Pirlot and Schneider, 1974). For two Edentates, Pirlot (1980) found fairly low progressions i.e. 145 (sloth) and 148 (armadillo). Bau-

chot and Stephan (1966) reported that placental Insectivores range from about 100 to 285. This gives a rough idea of how these independent groups have progressed above their respective standard levels; of course, no direct comparison of indices is possible since the standard levels are different.

The lack of extensive published data on the ecology and ethology of marsupials precludes a very detailed interpretation of their encephalization indices. However, some general informations extracted from semi-popularized publications may be meaningful in this context. I shall mainly draw from Ride's (1970) beautifully illustrated book on Australian mammals.

Vombatus, which reaches the highest figure on my scale, is reputed to have a fair degree of social life and a definite skill in the construction of underground dwellings. Its burrows are said to lend themselves to interchanges, sheltering as they do dense communities.

The kangaroos demonstrate a fairly great index variability (132 to 195) with the tammar *M. eugenii* at the lowest limit. The tammar is a small species but not the smallest on my list. It lives in «dense thickets in dry sclerophyll forest» (Ride, 1970) and can survive with very little water. Ecologically, it is entirely distinct from the other small wallabies and requires a particular stage of forest growth (Ride, 1970). It seems to be limited to a relatively precarious environment and its survival appears to be difficult in spite of its resistance to water shortages. The highest encephalization figures for the Macropodidae are exhibited by the smallest (*Potorous*) and the largest

TABLE III

*This table shows the percentage composition of the brain in each species.
The last line is to be considered separately since it is the sum of RH and PY, equivalent to RH in other studies*

Polyprotodonta															
%	Didelphidae									Peramelidae		Dasyuridae			
												Phascogalinae		Dasyurinae	
	Didelphinae					Microbiotheriinae				Peramelinae					
	Mar-mosa murina	Mar-mosa domina	Mar-mosa thylamys	Meta-chirops	Phi-lander	Didel-phus aurita	Didel-phus marsupialis	Didel-phus virginiana	Calu-romys	Pera-meles	Isodem	Ante-chinus	Smin-thopsis	Dosy-urus	Sarco-philus
Bo	5.39	7.78	6.83	6.57	8.42	6.47	9.00	8.91	4.21	8.94	8.59	7.51	7.46	6.77	7.50
N	24.08	21.63	21.73	24.59	22.32	20.82	22.24	19.73	22.45	16.66	17.26	16.53	20.47	25.99	24.63
RH	9.39	10.88	10.93	12.46	10.34	10.30	12.48	12.23	8.00	10.82	10.34	11.38	11.06	9.90	10.60
PY	4.65	4.48	6.08	4.58	5.83	4.68	5.82	5.52	4.13	7.04	5.80	5.73	5.29	4.55	4.42
S	1.06	0.77	0.64	0.78	0.76	0.68	0.93	0.88	0.93	0.86	1.32	0.85	0.82	0.71	0.67
Di	8.40	9.13	8.11	8.61	8.42	9.95	9.88	9.50	12.29	8.59	8.64	8.87	9.17	9.17	9.72
STR	3.35	2.49	2.56	2.36	2.70	2.14	3.11	2.50	2.57	2.98	3.03	3.33	3.05	2.89	3.26
Hi	8.63	8.12	9.69	8.34	9.23	8.34	8.26	7.28	8.39	10.57	11.69	8.07	9.29	6.64	5.43
M	9.70	9.26	9.99	6.41	7.30	7.45	5.84	6.58	8.15	5.84	6.50	10.47	8.68	6.56	6.10
C	13.42	12.39	13.25	13.35	13.91	15.92	13.13	16.51	16.00	15.94	15.03	12.98	11.91	16.12	16.50
O	11.36	12.44	8.52	11.21	9.93	11.31	8.72	9.90	11.61	10.35	10.20	13.45	11.97	9.69	10.03
Sz	0.57	0.63	1.67	0.74	0.84	1.93	0.59	0.47	1.27	1.41	1.60	0.83	0.83	1.01	1.14
RH PY }	14.01	15.36	17.02	17.05	16.16	14.98	18.22	17.75	12.11	17.86	16.14	17.10	16.34	14.44	15.01

TABLE III (b)

	Diprotodonta										
	Phalangeridae				Macropodidae						Vombatidae
	Phalangerinae	Phascolarctinae		Trichosurinae	Potoroinae	Macropodinae					
	Petaurus	Schoino- bates	Pseudo- cheirus	Tricho- surus	Potorus	Macro- pus Eugenii	Setonix	Thylo- gale	Macro- pus rufogriseus	Macro- pus gigonteus	Vombatus
Bo	2.42	2.50	1.80	2.54	5.40	2.65	2.14	2.65	1.61	1.16	2.62
N	30.96	30.85	30.91	36.57	32.99	37.70	34.07	40.75	41.33	46.45	45.82
RH	7.56	5.26	7.03	7.92	9.28	6.40	7.74	6.86	5.27	6.87	6.14
PY	3.70	1.66	2.04	2.17	3.85	3.11	2.53	1.83	1.04	1.33	2.44
S	0.66	0.69	0.81	0.54	0.69	0.72	0.59	0.41	0.32	0.47	0.56
Di	13.33	11.25	10.82	10.31	9.57	11.58	11.16	9.58	9.22	11.21	9.95
ST	4.12	4.02	2.73	4.34	3.25	3.10	2.87	3.01	2.14	2.87	3.72
Hi	6.63	5.52	8.68	5.53	11.04	8.30	8.32	8.21	6.54	5.58	5.49
M	4.66	6.94	6.40	4.20	4.12	4.57	5.87	4.02	4.25	3.00	2.72
C	15.07	19.61	16.61	15.20	12.23	14.00	14.98	14.65	18.17	13.37	13.48
O	9.70	10.47	11.91	9.73	6.67	7.06	9.08	7.50	9.53	6.81	5.86
Sz	1.19	1.23	1.08	0.92	0.91	0.81	0.65	0.56	0.58	0.88	1.20
PY } RH }	11.25	6.91	9.07	10.08	13.13	9.05	10.27	8.67	6.31	8.20	8.57

Table III (c)

Paucituberculata		
Caenolestidae		
	Lestoros	Caenolestes
Bo	10.23	8.94
N	17.49	19.80
RH	12.69	12.87
PY	7.27	7.74
S	0.85	0.70
D	7.52	8.10
ST	3.87	2.89
H	8.06	9.26
M	7.10	5.61
C	13.75	13.92
O	10.78	9.77
Sz	0.39	0.40
RH PY	19.96	20.61

(*M. giganteus*) in my sample. The potoroo, one of the rat-kangaroos, lives in dense vegetation (grass, heath, forest); it seems to be unable to escape its predators except by hiding and defends itself by restricting its movements to that particular cover which

is probably plentiful. The great grey kangaroo *M. giganteus* dwells «in eucalypt woodland and sclerophyll forest» (Ride, 1970), feeding on grass between the trees and shrubs. It stands in contrast to the red kangaroo (*M. rufus*) which is a species encountered in more open country, i.e. a pasture animal. The pademelon *Thylogale* seems to be little known and apparently lives under thick cover while the small quokka *Setonix*, which favours swampy thickets, is also reputed to be well adapted to dry conditions (Ride, 1970).

Among the Phalangerioidea (possums), there are extremes of encephalization. The highest figure is reached by the phalangerid *Trichosurus* a tree-dweller that is also said to be adjusted to life on the ground, which indicates a certain versatility of behaviour. In fact it is also adapted behaviourally to the vicinity of humans, living near suburban areas where it haunts roofs and eats cultivated plants (Ride, 1970).

TABLE IV

Constants for brain components equations in basal didelphinae

Components	B	k
Bulbus olfactorius	0.3705	0.7264
Neocortex	1.5184	0.5140
Rhinencephalon	1.0370	0.5848
Pyriform cortex	0.7294	0.5758
Septum	0.0604	0.5195
Diencephalon	0.9712	0.5748
Striatum	0.6305	0.4994
Hippocampus	1.0894	0.5187
Mesencephalon	1.2585	0.4360
Cerebellum	1.1388	0.5840
Medulla oblongata	1.2015	0.5090
Schizocortex	0.0499	0.5673
Rhinencephalon + Pyriform	1.2127	0.5815

For log equation $Y = B + kX$ where $Y = \log$ of component volume in mm^3 ; $B = \log b$ of allometric equation; $k = \text{growth exponent in the same equation}$; $X = \log$ of body-weight in gr.

TABLE V

Progression indices for total brains (encephalization)

<i>Caluromys lanatus</i>	134 (136)	<i>Trichosurus vulpecula</i>	188 (185)
<i>Antechinus flavipes</i>	89 (83)	<i>Potorous tridactylus</i>	191 (198)
<i>Sminthopsis murina</i>	93 (90)	<i>Setonix brachyurus</i>	164 (171)
<i>Dasyurus quoll</i>	133 (136)	<i>Thylogale billardieri</i>	166 (156)
<i>Sarcophilus harrisii</i>	128 (134)	<i>Macropus eugenii</i>	132 (138)
<i>Perameles nasuta</i>	102 (106)	<i>Macropus rufogriseus</i>	172 (183)
<i>Isodon obesulus</i>	81 (84)	<i>Macropus giganteus</i>	195 (208)
<i>Petaurus breviceps</i>	126 (127)	<i>Vombatus mitchelli</i>	221 (248)
<i>Petaurus norfolcensis</i>	123 (124)	<i>Caenolestes obscurus</i>	123 (121)
<i>Schoinobates volans</i>	78 (80)	<i>Lestoros inca</i>	101 (99)
<i>Pseudocheirus peregrinus</i>	124 (127)		

Table V is based primarily on the basal regression $Y = 2.1446 + 0.5315 X$ estimated from the 8 Didelphinae in Tables I and II, as explained in text. These are the only didelphines for which I had brains for sectioning, from the beginning of this work. In a later check of the literature (Moeller, 1973), I found BrW/BoW figures for one more genus-species, *Lutreolina crassicaudata* (BoW = 1355 gr; BrW = 5.0 gr). This addition is interesting although I cannot use it for detailed analysis since Moeller (1973) did not publish data on the components. For the sake of improving the basal correlation, I reworked the BrW/BoW didelphine correlation with *Lutreolina* and obtained $Y = 2.1698 + 0.5188 X$. The progression indices with respect to this new basal line are shown in brackets in Table V. The average difference in the PI values is less than 4 % which is perhaps not negligible but does not change the relative positions.

TABLE VI

This Table presents the progression indices (PI) of the brain components for all species other than Didelphinae. The index for the pooled RH + PY has again been estimated separately

Polyprotodonta							
PI	Didelphidae	Peramelidae		Dasyuridae			
	Microbiotheriinae	Peramelinae		Phascogalinae		Dasyurinae	
	Caluromys	Pera- meles	Isoodon	Ante- chinus	Smin- thopsis	Dasy- urus	Sorco- philus
Bo	96	131	110	184	252	145	99
N	131	77	70	60	80	166	141
RH	92	92	77	98	111	119	101
PY	102	131	94	104	110	118	92
S	152	110	148	88	90	126	106
Di	175	92	81	92	110	137	117
STR	127	118	105	99	95	158	163
Hi	128	128	123	78	97	110	80
M	139	85	83	91	74	129	124
C	147	109	89	89	95	154	125
O	145	103	89	104	98	133	124
Sz	159	133	132	74	85	133	121
RH } PY }	95	105	82	100	111	118	98

TABLE VI (b)

Diprotodontia											
P I	Phalangeridae				Macropodidae						Vombatidae
	Phalangerinae	Phascolarctinae		Trichosurinae	Potoroinae	Macropodinae					Vombatus
	Petaurus	Schoino- bates	Pseudo- cheirus	Tricho- surus	Potorus	Macro- pus Eugenii	Setonix	Thylo- gale	Macro- pus rufo- griseus	Macro- pus gigon- teus	
BO	58	26	34	50	148	35	35	37	25	18	52
N	158	108	168	261	295	199	221	282	335	435	507
RH	81	33	73	99	151	57	68	75	66	96	104
PY	85	23	45	59	136	61	61	44	29	41	91
S	95	66	121	106	171	104	106	77	71	120	168
Di	176	89	139	162	194	131	156	135	148	202	215
ST	176	120	126	267	249	142	162	184	154	240	366
HI	89	50	123	103	256	113	140	146	136	134	156
M	65	77	106	99	117	83	130	103	131	109	116
C	129	99	136	152	158	100	133	129	182	150	181
O	106	79	129	149	128	80	127	113	168	139	141
Sz	136	86	121	128	144	81	80	71	83	142	234
RH PY	82	30	64	86	146	58	78	66	55	79	100

TABLE VI (c)

Paucituberculata		
Caenolestidae		
P I	Lestoros	Caenolestes
BO	311	328
N	75	103
RH	132	162
PY	158	203
S	102	104
D	94	121
ST	135	123
H	92	128
M	72	69
C	113	139
O	98	108
Sz	41	52
RH } PY }	140	175

This may be a richer environment than ordinary wilderness and one that it exploits more than any other phalangeroid. Of the Petauridae (gliders proper and ring-tail), which are all less encephalized than *Trichosurus*, the lowest ranking is *Schoinobates* (greater glider) which shows a regressive index. This extraordinary result should be confirmed by more data before being considered established. Ride (1970) notes that each of the genera of the gliding possums is more closely related to a non-gliding possum than it is to

the other gliders». The genera *Petaurus* (sugar glider) and *Pseudocheirus* (ringtail) demonstrate an equal, moderately progressive encephalization. *Schoinobates* is reputed to feed on eucalypt leaves only, to be sedentary, to glide less frequently and to manoeuvre less skilfully than some *Petaurus* types at least (Ride, 1970). Further, it is silent and perhaps olfactory in its habits (Tyndale-Biscoe, 1973) while *Petaurus* is vocally expressive, feeds on various blossoms, on the sap of branches and on insects (Ride, 1970). It must also be noted that *Pseudocheirus* has a social or semi-social behaviour while *Schoinobates* tends to be solitary (Ride, 1970; Tyndale-Biscoe, 1973). Thus, if one compares *Schoinobates* behaviourally with the other two genera, it seems to be somewhat inferior on the psychic level; this may correspond to its lesser encephalic development. *Petaurus* and *Pseudocheirus* may be significantly more versatile, less specialized in their habits. For general habitats, the three genera choose woodlands and forests, *Pseudocheirus* sometimes showing predilection for areas along water-courses and swamps (Ride, 1970). It is possible that *Schoinobates* is more strictly arboreal, more canopy-dwelling than the others.

TABLE VII

Progression indices with respect to Phascogalinae and Peramelidae

<i>Dasyurus</i>	139	<i>Potorous</i>	209
<i>Sarcophilus</i>	138	<i>Setonix</i>	179
<i>P. breviceps</i>	140	<i>Thylogale</i>	179
<i>P. norfolcensis</i>	136	<i>M. eugenii</i>	143
<i>Schoinobates</i>	85	<i>M. rufogriseus</i>	186
<i>Pseudocheirus</i>	136	<i>M. giganteus</i>	209
<i>Trichosurus</i>	206	<i>Vombatus</i>	248

Australian species taken from Table I. The basal equation used here is $Y = 2.0876 + 0.5373 X$ with $r = 0.99$. Compare with Table V.

Among the Dasyuridae, two groups are easily distinguished. *Dasyurus* and *Sarcophilus* form a subfamily of carnivores (Dasyurinae) that inhabit dry sclerophyll forests and are of relatively large size in comparison with the other subfamily (Phascogalinae). The latter comprises *Antechinus* and *Sminthopsis*, two small mouselike genera, more or less ubiquitous (several species from dry and wet forests, savannah, arid areas and various altitudes), with a diet similar to that of placental Insectivores. Between these two groups, there is a clear difference in encephalization. The Phascogalinae *Antechinus* and *Sminthopsis* stand at a definitely lower level within the Dasyuridae. I shall discuss this further in the conclusions.

The last Australian group in my material is the family Peramelidae. Two genera of bandicoots represent here that family, *Isoodon* and *Perameles*, and both show a low encephalization (81 and 103). Both live in a variety of environments with good vegetal cover and seem to be fairly eclectic in their diets again, more or less like primitive Insectivores, feeding on insects and other invertebrates, with *Perameles* also taking berries. *Isoodon* is capable of digging. Both are nocturnal, somewhat secretive, solitary and even strongly territorial in their individual behaviour (Ride, 1970). Note that my findings on the relatively low position of the Phascogalinae and Peramelidae agree with Moeller's (1973) results.

In the American marsupials, one may notice that *Caluromys*, if taken as a separate didelphine (see Appendix A), is relatively progressive with an index of 134. Even if Reig's suggestion of placing that genus within the

Microbiotheriinae with *Dromiciops* must be abandoned, it remains true, according to Reig again endorsed by Kirsch (1977) that it is a distinct form. The encephalization coefficient confirms that view.

Finally, the last American genera on hand are the rare Caenolestidae *Lestoros* and *Caenolestes*. To me, the difference in encephalization between the two is intriguing, albeit small (101 and 123). Both animals are lowly encephalized but *Lestoros* is not progressive at all whereas *Caenolestes* is moderately so. I lack good information on the life-habits of those animals.

B. COMPONENT PROGRESSION

I will start again from the highest ranking genus, using the size of the neocortex as the main criterion. I shall not comment in detail on every index because the comparative significance of some indices cannot be properly evaluated until more data on the life-habits of these species become available.

Vombatus occupies, as expected, the top of the scale for neocortical development. This agrees with the image of an industrious and social animal. Its olfactory components are either regressive (BO) or on a level with the basal type (RH + PY). Among the high figures for that species, the striatum is exceptional (366), which may indicate retention of a primitive feature. This is true also of the schizocortex (234), and this probably has the same implication. Note that the cerebellum also has the highest index of all the species studied, i.e. it is more progressive in the terrestrial-fossorial vombat than in the acrobatic phalangerids. This raises (and probably answers in part) the question of the

significance of a large cerebellum, i.e. whether it simply follows neocortical development or indicates an adaptation to complex locomotion (this has been suggested for mammals with an arboreal or aquatic life). As already supported by our work on bats (Stephan and PirLOT, 1970; PirLOT and Pottier, 1977), the correlation with the neocortex seems to be of primary importance.

Among kangaroos (Macropodinae), the range of variation is fairly broad for the neocortex (199 to 435). Again, the tammar (199) may appear as a specialized and not too successful species, when compared with other types of *Macropus*. All have strongly regressive olfactory components which is of course not surprising for herbivorous animals. The cerebellum is remarkably low (100) in *M. eugenii*, most probably because it keeps pace with the neocortex, unless the explanation is that the locomotion of that species is very different from that of the other kangaroos; however, to my knowledge, there are no data to indicate this. As to the schizocortex of the Macropodinae on my list, its variability may merely reflect the great difficulty I encountered in determining its limits.

Quite exceptional is *Potorous*. As seen above, its encephalization already indicates a relatively big-brained animal. But the detail shows a) that its neocortex is clearly less developed (295) than that of large kangaroos, while more developed than that of the three small ones on my list; b) that its olfactory system is enormously more progressive than that of any of my Macropodinae; c) that its striatum and its limbic system (septum, hippocampus and schizocortex) are also

exceptionally large. Without offering any explanation for it, I cannot help feeling that the potoroine «originality» is both undeniable and worth further investigations (neurological and ethological). The «primitive» components in *Potorous* seem to predominate and account for the size of its brain: but that the brain itself should be so voluminous under such circumstances remains puzzling.

Among the Phalangeridae, *Trichosurus* looks superior both in neocortical development and in encephalization. Its cerebellum is also the most progressive of the group. Its striatum and schizocortex are also remarkably large; in some animals, this is known to be incompatible with a large neocortex. The latter is minimal in *Schoinobates* and so is its cerebellum, a rather unexpected observation in an arboreal animal. This is further demonstration of the importance of the neocortical-cerebellar correlation. In all four genera, the olfactory components are strongly regressive, even in the reputed «olfactory» genus *Schoinobates* (Tyndale-Biscoe, 1973).

The Dasyuridae also divide into two very distinct groups on the basis of neocortical progression. The Phascogalinae stand far behind the Dasyurinae and appear to be extraordinarily regressive. The cerebellum correlates in the same way and the limbic system, especially the schizocortex, seems to go along with the higher centers; it is underdeveloped in the Phascogalinae but progressive in the Dasyurinae. In all, the olfactory structures have a relatively high index, which is to be expected from carnivores-insectivores. The striatum is also strikingly more progressive in the Dasyurinae than in

the Phalangerinae, possibly a retained ancient feature.

The two Peramelidae follow the pattern of the Phascogalinae as far as neocortex and olfactory structures are concerned, but their cerebellum is somewhat more progressive as is their limbic system.

Caluromys again distinguishes itself from the rest of the Didelphinae by its large neocortex and cerebellum and by the progressiveness of its limbic structures. Its diencephalon seems to be exceptionally large: its olfactory system is equal to that of the main didelphine group.

The Caenolestidae have low neocorticalization but a highly progressive olfactory system with the most extraordinary bulbi. One would expect the sniffing ability of those two genera to be most remarkable. Their other components are in general only moderately progressive.

CONCLUSION

On the basis of the data presented here and the accompanying information, I wish to offer a few conclusive remarks on two distinct aspects of quantitative brain development in marsupials.

BRAIN-BEHAVIOUR RELATIONSHIPS

a) The discrimination between the various species on the basis of encephalization alone is well marked between the extremes. At the lowest level, we find the Phascogalinae (of the family Dasyuridae) and the Peramelidae, plus one exceptional glider (*Schoinobates*) and also *Lestoros* of the Caenolestidae. These animals are likely to show similarity of insectivorous — carnivorous habits, being also

mainly walking animals (except the problematic *Schoinobates*) but quite capable of climbing, as are many other small mammals. One may refer to their life-habits as rather unspecialized and primitive. At the highest level, we see the vegetarian *Vombatus* with its elaborate social life and immediately below it all the browsing and grazing Macropodids, as well as the Phalangerids other than *Schoinobates*, among whom there probably exist various vegetarian, probably frugivorous diets, distinct from the restricted leaf-feeding observed in the regressive and specialized *Schoinobates*.

b) The olfactory structures as they appear in Didelphinae (see percentages in Table III) characterize an ancient and basic, if not ancestral, type of marsupial with a primarily insectivorous but also partly carnivorous diet. This reminds one of the basal Insectivores as defined by Stephan (1967). A similarly high degree of olfaction is suspected in the peramelids which are said to have analogous feeding habits and again in the more carnivorous dasyurids, as one can expect. A lesser development is observed in the olfactory components of *Caluromys*, which supports the separation of that genus from the other didelphids, and also in *Petaurus* and *Trichosurus* and even more in *Pseudoscheirus* and *Schoinobates*. In the light of this finding, despite its reputation, the olfactory behaviour of *Schoinobates* deserves re-examination. Although *Schoinobates* is often seen sniffing intensively, since its exclusive food is made of strongly scented eucalypt leaves, it perhaps does not take a large olfactory system to find it: this could perhaps be viewed as an adaptation for a *bad*, not a good, sniffer!

The macropodids and the wombat have poorly developed olfactory structures with the exception of *Potorous* which is endowed with relatively large bulbi and rhinencephalon, an unexplained feature due to insufficient ethological reports. Note that, as emphasized elsewhere (Pirlot, 1974, p. 154), other features than mere volume may play an important part in this kind of morpho-functional relationship (e.g. number and pattern of synapses, membrane properties, submicroscopic organization of neuron bodies, etc).

c) Both the percentages and the progression indices show the neocortical superiority of the wombats and the large kangaroos. Besides, the potoroo also ranks high among the macropodids. As already said, one may stress the industriousness and social life of the wombat. As to the kangaroos in general, they are social and visual animals to various degrees (not unlike placental Ungulates in their habits) which probably calls for moderately high neocorticalization. The gliders and the carnivorous dasyurids are definitely lower, again as expected. *Caluromys* stands out among the didelphids for its larger neocortex, another reason for keeping it separate. Both caenolestids are unprogressive, which points to their primitive status on the basis of the neocortex; they may be even specialized in their olfaction, possessing as they do oversized bulbi.

d) My observations on the cerebellum are inconclusive except for the fact that there is a general correlation between that component and the neocortex. The highest cerebellar index is found for a large *hopping* kangaroo; it is practically equal to that of the *fossorial* wombat. Further, no clear indication of cerebellar development in

relation to arboreal life was found, although *Trichosorus* and *Dasyurus*, which are good climbers, do have a fair size cerebellum. Among the hoppers, the situation is uncertain with some high (182, 158) and some low (100) figures. *Caluromys* and the caenolestids stand at intermediate levels.

MARSUPIAL BRAINS AND MARSUPIAL PHYLOGENY

I will not become involved here in the long and complex debate over marsupial origin and early phylogenetic branching. However, I want to point out some elements of discussion taken from this study, which may prove useful to marsupialologists attempting to bring together the available morphological, physiological and behavioural data.

The Didelphinae, as a group, remain a valid reference set even if they cannot be taken to represent the ancestral form for the whole subclass. Their established similarity with the oldest known marsupials and their overall biological homogeneity combined with a sufficient size-range and a generalized, little-specialized aspect, constitute a justification of this (Appendix A). Rat-like, long-nosed, walking and climbing with subequal limbs, olfactory, mainly nocturnal, insectivorous — carnivorous and occasionally omnivorous with relatively low encephalization and neocorticalization, most of them are probably not far from an ancestor that may also bear some resemblance to the ancestral Insectivore imagined for the placentals.

That does not designate them as unquestionable ancestors to all marsupials, as already said. To the arguments presented by other authors with

respect to various osteological and serological features, I can add some observations related to brain organization. First of all, the fact that at least two groups of Australian marsupials (Phascogalinae, Peramelidae) are clearly regressive in comparison with the didelphids, carries important implications. The idea of an independent branching off near the marsupial point of origin appears a serious possibility. The significance of that regressiveness, however, remains difficult to evaluate as one observes the low index *also* found in *Schoinobates* whose taxonomic place among more advanced gliders is, as far as I know, undisputed. In any case, the Phascogalinae and the Peramelidae seem to be relatively close to each other with respect to encephalic organization (see Appendix A).

A similar problem arises in the case of the South American caenolestids that stand low on the scale of evolution yet rank relatively high in specialization. How old is that family? It could be a *very* early offshoot from the main marsupial stem.

One of the main problems related to marsupial phylogeny is to elucidate the relationships between American and Australian groups. Does one derive from the other or are they independent? Classically, as we have seen, and mainly on paleontological grounds, it has been assumed that they have a common ancestor on the American continent and that a marsupial dispersal took place either through a Northern (Arctic) passage or along a Southern (Antarctic) route. But the idea of two independent evolutions dating back to a premarsupial separation can also be considered in broad agreement with Tyndale-Biscoe (1973, p. 35).

My present data can be used to start the discussion afresh. Indeed, the fact that the Phascogalinae and the Peramelidae are both regressive in their brain development may lead one to think (a) either that they represent the ancestral type of *all marsupials* better than the Didelphidae; (b) or that they represent at least the ancestral type of *the Australian marsupials*, having only a premarsupial relation with the American fauna. Hypothesis (a) is not supported by any paleontological or other evidence that I know of. Furthermore, one should examine if the Phascogalinae and the Peramelidae possess the «generalized» features found in Didelphinae, as described above. Hypothesis (b) is less risky and I sympathize with it to a large extent. However, more paleontological data to support it would be welcome. On the other hand, one may look at the brain progression by referring all the Australian species to the Phascogalinae and/or Peramelidae taken as the «basal Australian standard». I hope that this will be done some day with a large sample from these two groups (and a sample showing both the homogeneity and the size-range that is required, as stressed in Appendix A). In the meantime, I have tentatively estimated a basal regression with the few Phascogalinae and Peramelidae present in my list. The equation almost coincides with that obtained for the Didelphinae and, as expected, the progression indices for the Australian species, when based on that equation, hardly differ from those obtained with the didelphine basal line (Table VII). Thus, I come back to the conclusion that the didelphine reference remains as good as any other one for the time being, especially because it agrees with significant paleontological data. At

the same time, I believe that the low indices found here for the Phascogalinae and the Peramelidae, especially if they should be confirmed by further studies, point to the possibility or even to the likelihood of an early separation of both lines, that is the American and Australian stocks could have been separated *since premarsupial times*, undergoing thereafter independent evolutions. In that perspective, the argument over a Northern or a Southern migrations in the late Mesozoic loses much of its appeal.

APPENDIX A

1) The choice of the Didelphinae as the reference basal group in this study may be disputed. My reasons for that decision are as follows. At the time of writing, Clemens' papers (1968, 1977) on the phylogeny of marsupials appear the best available studies on that subject. From them, it seems most likely that «Differentiation of the Marsupialia in the early Late Cretaceous was followed by a major adaptive radiation in North America. All these Cretaceous lineages except the didelphid, represented by the species of *Alphodon*, became extinct at the end of the period» (Clemens, 1968, p. 16). From the «oldest record of unequivocal marsupials [...] from the Milk River Formation of Alberta, Canada» (Clemens, 1977, p. 58), the evidence is toward the didelphids, which are still extant. This does not, of course, mean that the present didelphids are the perfect image of a «universal» marsupial ancestor. But, in the words of an authority (personal communication — J. Kirsch) on the subject, «while I think it technically true that didelphids are probably not the basal group, they are the best approximation». I have taken this balanced view as a sound starting point before working out the final plan for this paper. My own results, fragmentary as they may be, in fact agree with it, due consideration being given to the hypothesis of an early independent origin for the Australian stock, as explained above. It must be remembered that part of the value of the basal group method as proposed by Stephan (1967) is to combine the advantage of primitiveness in brain structure with the advantage of «ancientness» in time, by which the probability of the

reference standard being close to the true ancestral type is enhanced. As a matter of fact, in two international symposia on vertebrate evolution, I have heard at least as many expression of scepticism on the basal character of Insectivores in the case of placentals as I have read criticisms on the basal value of didelphids in the case of marsupials.

2) One might be tempted to apply a similar logic to the entire class of mammals and propose to take the Insectivores as the basal group for both the placentals and the marsupials. That, I believe, would be going too far. I can support the choice of the didelphids as standard marsupials for both the American and the Australian species, on the paleontological grounds indicated. But taking the Insectivores as representing *the archetype of all mammals* would be a much more audacious step, and one that is unwarranted by what we now know about our hypothetical Therapsid ancestor.

3) In the discussion, I have tentatively taken together my Phascogalinae and Peramelidae as the Australian basal set. I have done this mainly on the basis of my encephalization indices. But Dr. J. Kirsch informs me (personal communication) on the basis of his own protein and dentition studies that «the small dasyurids and the peramelids intergrade completely». Our views thus agree.

4) As a reference group in using the progression index method, one must select a set of species that satisfies two prerequisites at least. a) The set must provide a broad range of sizes, for example with a ratio 1 to 25 or more. b) The set must involve as much taxonomic, morphological and ecological homogeneity as possible, along with the «generality and primitiveness» already referred to. The basal group chosen here is composed of species belonging to a single *subfamily*, and showing a 1 to 100 size ratio between the extremes.

5) The inclusion or exclusion of *Caluromys* as a basal genus raised a problem. Following Reig (1955, in Kirsch 1977) I should exclude it as one of the Microbiotheriinae. Later (Kirsch and Calaby, 1977) it appears that Reig and Kirsch have agreed that it belongs to the Didelphinae «though undoubtedly distinct from typical didelphines» (Kirsch 1977, p. 46). Based on this opinion, I have decided to exclude it from the basal set used for the calculation of the progression indices. However, for curiosity's sake, I have estimated a second

time the progression indices for encephalization on an equation including *Caluromys*. That equation is $Y = 2.1587 + 0.5304 X$ with $r = 0.9853$; it is hardly different from the one worked out without *Caluromys*. This change does not significantly affect the indices as the reader may verify for himself.

APPENDIX B

The delimitation of neuronal fields in this paper is based almost entirely on the model afforded by the excellent atlas *The Brain of the Opossum* by Cruz, E. O. and C. R. Miranda (1968). I have experienced some difficulties with two brain components: a) the septum which no work known to me identifies to my satisfaction: I have confined my limiting lines to the most obvious septal nuclei area and this may put that component at its lowest possible value in my tables; b) the schizocortex which most atlases and descriptive papers do not even bother to mention. A small amount of that component is identifiable in all marsupials. Johnson (1977) shows it in the subicular region under the name of entorhinal cortex. Another portion of it can be seen in certain cross sections near the pyriform cortex, laterally. Both parts may be seen to join caudally. The limits are always difficult to trace with any

accuracy, particularly in marsupials, although in placentals too they vary according to the author and they are often not very reliable. Because of that uncertainty, the schizocortex of *Didelphis* was not distinguished in a previous paper (Pirlot and Nelson, 1978). In this work, I have restricted it to what looked to me quite undisputable.

APPENDIX C

It would be a mistake to expect, as some authors may be tempted to do, strictly characteristic values for the BrW/RoW ratios for species, genera or families. Species-specific weight ratios are just impossible to estimate unless one uses very large samples, which most students do not have at their disposal. In marsupials, the intra-group variability of the BrW/BoW ratio is considerable, probably much greater than it would be in placentals because of the peculiar marsupial metabolism. That is a drawback that no dogmatic claim can avoid. As far as the preparation of the brains and the «normalness» of the body-weights are concerned in the present material, I have checked with the collectors that the collecting procedures for all specimens do not entail any risk of seriously biasing recording.

ABRÉGÉ — Étude quantitative sur le cerveau des marsupiaux dans une perspective éco-éthologique. Utilisant une méthode volumétrique et l'équation classique d'allométrie, l'auteur étudie l'éventualité d'une relation entre la dimension du cerveau, et l'écologie ou éthologie de 29 espèces de marsupiaux sud-américains et australiens. Il montre qu'il existe une relation entre les volumes cérébraux et néo-corticaux, et la position taxonomique caractérisée par des mœurs définies. L'évolution des marsupiaux est ainsi remise en lumière ce qui montre que, les didelphidés actuels étant sans doute les plus proches du type ancestral, d'autres groupes tels que les dasyuridés et péramelidés comprennent des formes encore plus primitives. Ceci suggère que la séparation des marsupiaux américains et australiens fut très précoce, que leur évolution se déroula indépendamment depuis très longtemps, peut-être depuis une époque immédiatement prémarsupiale.

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